

Lesson 36: Tangents, Normals & Rates of Change — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. $y = x^2$. Find dy/dx .
2. $y = x^2$, gradient at $x = 4$?
3. $y = x^3$, gradient at $x = 2$?
4. Equation of a line through (1,2) with gradient 3?
5. Normal gradient if the tangent gradient is 2?
6. $y = x^2$, gradient at $x = -3$?

Section 2 — Core practice

7. $y = x^2$. Find the tangent at $x = 3$.
8. $y = x^2 - 4x$. Find dy/dx and the gradient at $x = 1$.
9. Normal gradient where the tangent gradient is 5?
10. $y = x^3$. Find the gradient at $x = 1$.
11. $y = x^2$. Find the tangent at $x = 1$.

12. Find the equation of the tangent to $y = x^2$ at $x = 2$.

Section 3 — Challenge & exam-style

13. $y = x^2 - 4x$. Find the equation of the tangent at $x = 1$.

14. $y = x^2 - 4x$. Find the equation of the normal at $x = 1$.

15. $y = x^3 - 3x$. Find the tangent at $x = 2$.

16. The volume $V = t^3$ is changing with time. Find the rate dV/dt at $t = 2$.